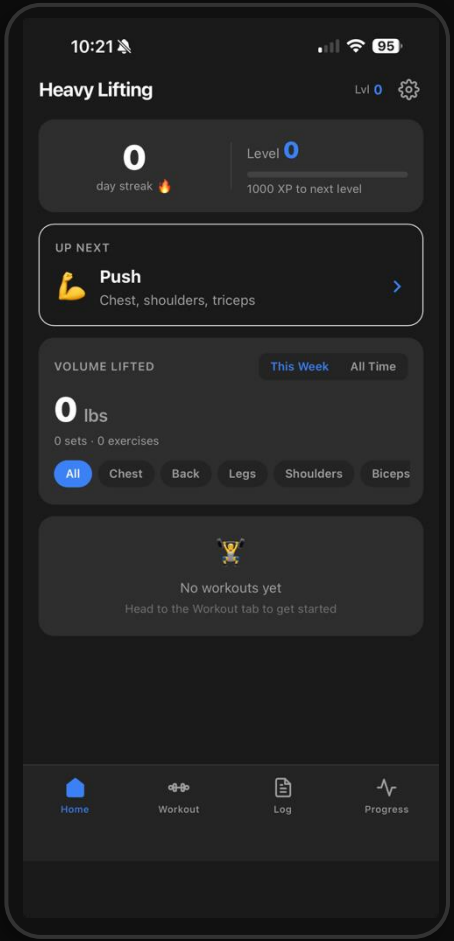
Heavy Lifting

A WORKOUT APP BUILT WITH AI

From idea to full-stack PWA — powered by Claude Code



Mason Hicks · April 2026

WHAT IS HEAVY LIFTING?

A full-stack fitness PWA built entirely with AI

🏋️ Smart Workouts

AI-generated PPL programs personalized to your gym equipment and history

📊 Progress Tracking

Lift charts, volume trends, PR history, and XP gamification

⚡ Works Offline

Full PWA with service worker — no internet required after install

🤖 Claude Integration

Optional Claude API for enhanced exercise swaps and weight recommendations

UNDER THE HOOD

Tech Stack

FRAMEWORK **React 18 + Vite**	Single-page app, no router — tabs via useState
STYLING **Tailwind CSS v3**	Custom design tokens, dark theme, CSS vars
DATABASE **Dexie.js (IndexedDB)**	Live queries, offline-first, persistent storage
CHARTS **Recharts**	LineChart + BarChart for lift/volume history
AI **Anthropic Claude API**	Exercise swaps + weight suggestions (optional)
PWA **vite-plugin-pwa + Workbox**	Service worker, offline cache, iOS install

THE DEVELOPMENT PROCESS

Built with Claude Code

Claude Code served as the primary developer throughout this entire project — from initial architecture to final polish.

01

Architecture Planning

Designed the full file structure, data models, and state machine before writing a single line

02

Feature Development

Wrote all React components, Dexie queries, and utility functions across 40+ files

03

AI Integration

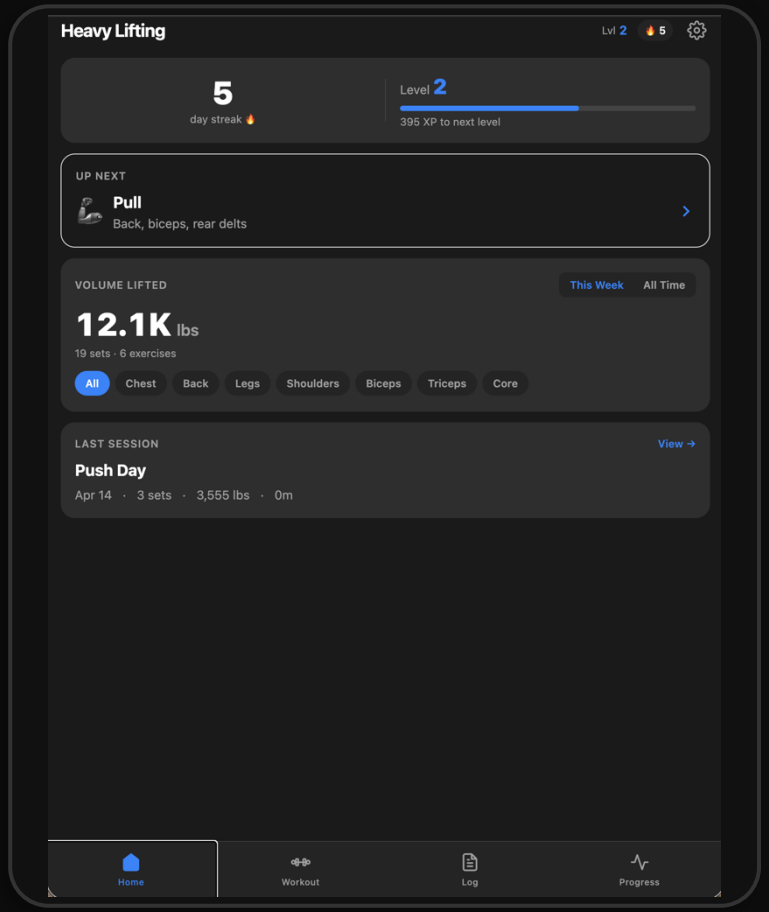
Implemented the workout generator logic + Claude API client with offline fallbacks

04

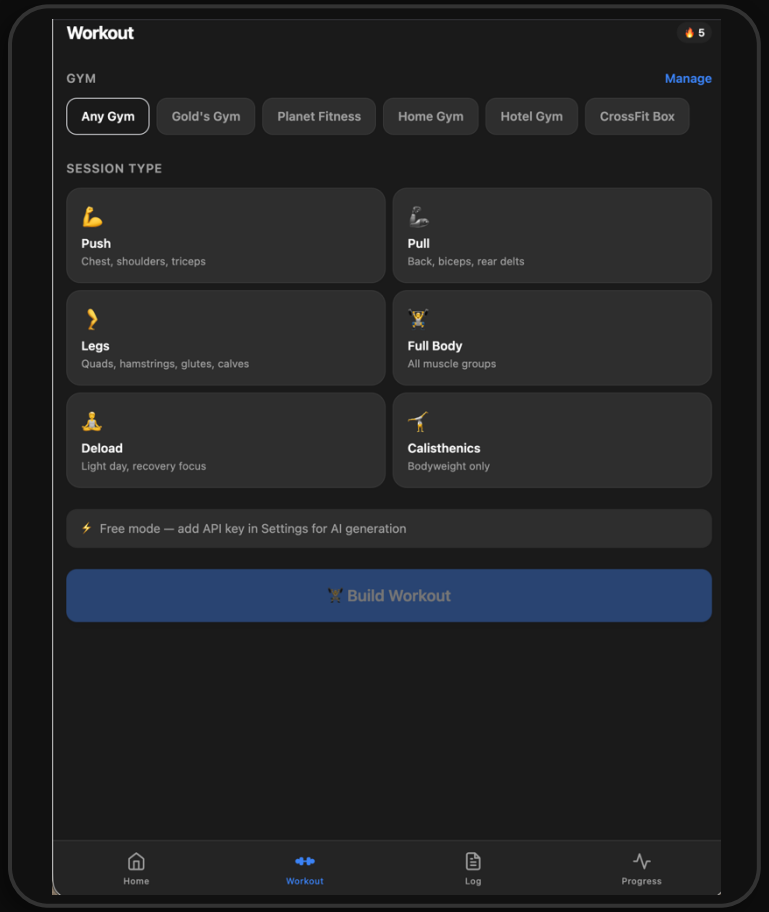
Testing & QA

21 Vitest tests, build verification, mobile viewport checks, iOS Safari fixes

THE APP IN ACTION

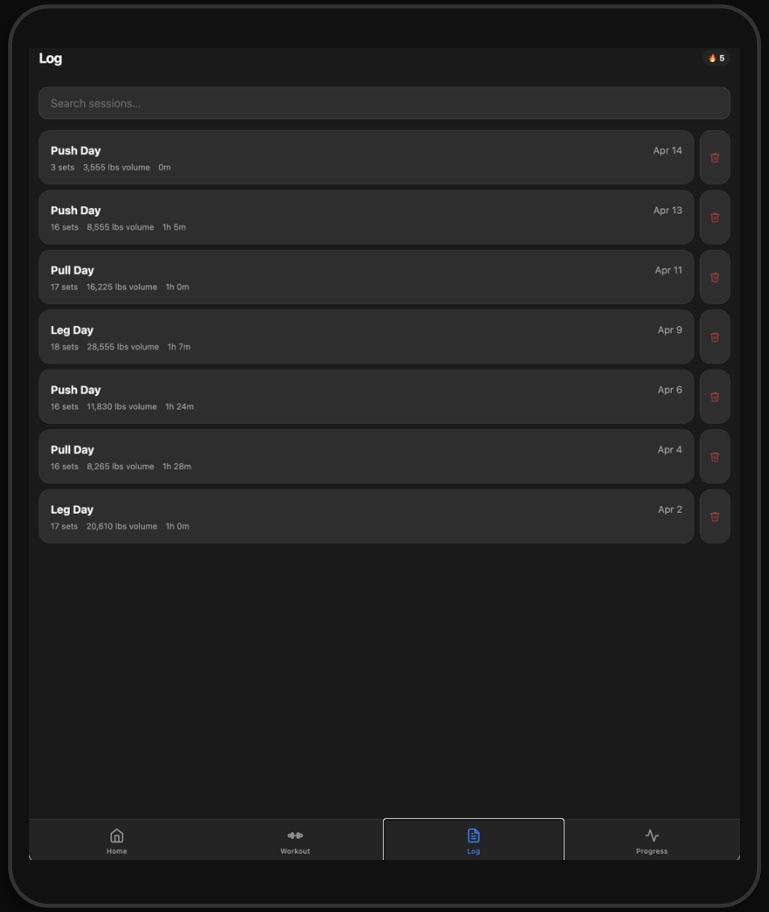


HOME DASHBOARD

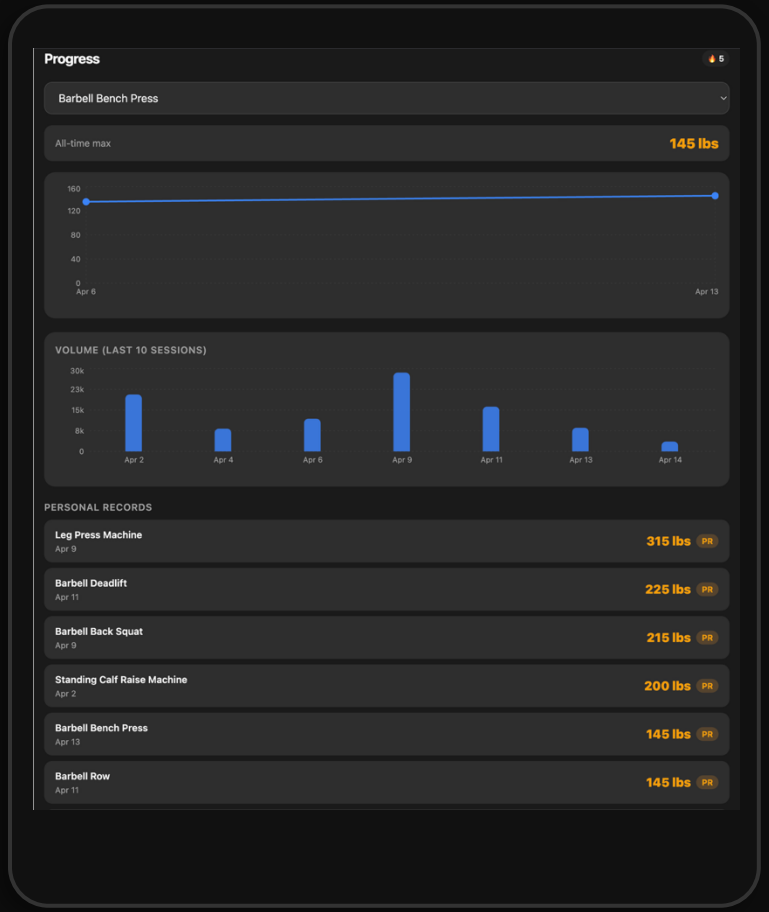


WORKOUT PLANNER

HISTORY & ANALYTICS



SESSION LOG



PROGRESS CHARTS

BY THE NUMBERS

What Claude Code built

40+

Source Files

Components, hooks, utils, tests

130

Exercises

Full exercise library with metadata

21

Vitest Tests

Workout generator fully covered

6

Weeks

Seed data generated for demos

733KB

Bundle Size

Production build

100%

Offline-first

PWA with service worker

AI INSIDE THE APP

Claude powers the workout intelligence

Pure-Logic Workout Generator

- PPL rotation (Push/Pull/Legs) — no API required
- Selects exercises by gym equipment + history
- Auto-deload when training 5+ consecutive days
- Muscle diversity: max 2 exercises per muscle group

Claude API Integration (Optional)

- AI-powered exercise swap suggestions
- Weight recommendations with ±15% cap from base
- Contextual session notes and coaching cues
- Full offline fallback — works without API key

Smart Weight Recommendations

- Layer 1: Last logged weight or exercise default
- Layer 2: Ease ratings (too-easy +5%, too-hard −10%)
- Layer 3: AI overlay capped at ±15% from base
- 3× "just right" = automatic progressive overload

KEY TAKEAWAYS

What this project demonstrates

🚀 AI Accelerates Full-Stack Development

A complete PWA with 40+ files, 130 exercises, 21 tests, and a gamification system — built in sessions rather than months.

🧱 Incremental, Always-Deployable Builds

Each session left the app in a working state. Claude Code followed a strict "finish one feature completely" philosophy before moving to the next.

🔒 AI Without Lock-In

The entire app works offline without an API key. Claude enhances the experience but every feature has a deterministic fallback.

📐 Architecture First, Code Second

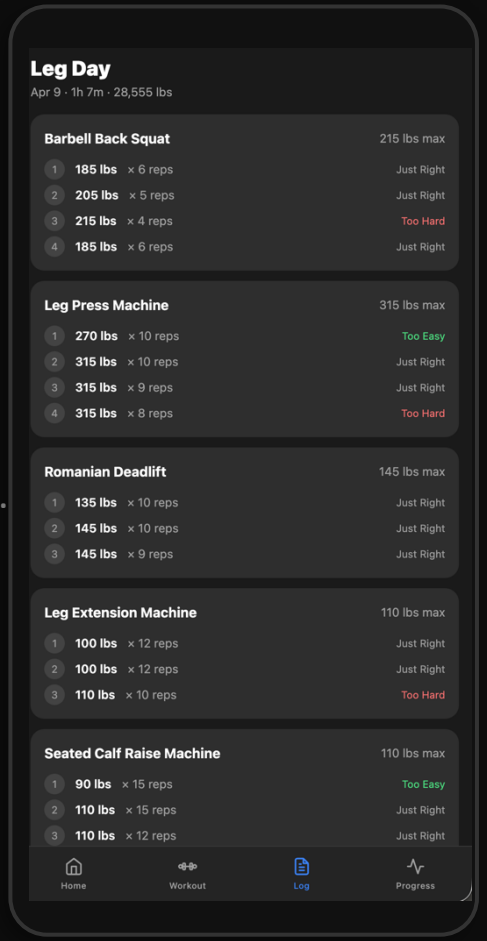
Claude designed data models, state machines, and component boundaries before writing code — leading to a clean, maintainable codebase.

Heavy Lifting

Built with Claude Code — from zero to production in days, not months.

React 18 | Tailwind | Dexie.js

Recharts | PWA | Claude API



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